

Analysis 1: Aug-Nov 2026

Definition: $\langle b_k : k \geq 1 \rangle$ is said to be a subsequence of $\langle a_n : n \geq 1 \rangle$ if there exists a strictly increasing sequence $1 \leq n_1 < n_2 < \dots < n_m < n_{m+1} < \dots$ such that $b_k = a_{n_k} \quad \forall k \geq 1$.

Definition: θ is said to be a limit point of a sequence $\langle a_n : n \geq 1 \rangle$ if given any $J \geq 1$ and $\epsilon > 0$, $\exists m \geq J$ such that

$$(\theta - \epsilon) < a_m < (\theta + \epsilon).$$

For $c \in \mathbb{R}$ and $\epsilon > 0$, the interval $(c - \epsilon, c + \epsilon)$ is called ϵ neighbourhood of c . For $c, d \in \mathbb{R}$, $c < d$, the interval $(c, d) \subseteq \mathbb{R}$ is called an *Open Interval* and the interval $[c, d] \subseteq \mathbb{R}$ is called a *Closed Interval*.

Exercises set #2

- (1) Suppose $\langle a_n : n \geq 1 \rangle$ is a bounded sequence, $\limsup_n a_n = \alpha$. Show that given any $J \geq 1$ and $\epsilon > 0$, $\exists m \geq J$ such that

$$(\alpha - \epsilon) < a_m < (\alpha + \epsilon).$$

The equation above is also written as $a_m \in (\alpha - \epsilon, \alpha + \epsilon)$.

- (2) Show that there exists a subsequence $\langle b_k : k \geq 1 \rangle$ of $\langle a_n : n \geq 1 \rangle$ such that $b_k \rightarrow \alpha$.
- (3) Show that there exists a subsequence $\langle c_k : k \geq 1 \rangle$ of $\langle a_n : n \geq 1 \rangle$ such that $c_k \rightarrow \beta = \liminf_n a_n$.
- (4) Let θ be a limit point of a sequence, $\limsup_n a_n = \alpha$. Show that exists a subsequence $\langle d_k : k \geq 1 \rangle$ of $\langle a_n : n \geq 1 \rangle$ such that $d_k \rightarrow \beta = \theta$.
- (5) Give an example of a bounded sequence $\langle a_n : n \geq 1 \rangle$ that has exactly two (distinct) limit points.
- (6) Give an example of a bounded sequence $\langle a_n : n \geq 1 \rangle$ that has infinitely many limit points.
- (7) Let $\langle a_n : n \geq 1 \rangle$ be a sequence such that $\lim_n a_n = u$.
- (i) If $u \in (c, d)$, show that $\exists m \in \mathbb{N}$ such that $a_n \in (c, d)$ for all $n \geq m$.
 - (ii) If $\exists m \in \mathbb{N}$ such that $a_n \in [c, d]$ for all $n \geq m$ then $u \in [c, d]$.